MIL-C-47011(MI)
19 April 1974
SUPERSEDING
MIS-13098
13 April 1965

MILITARY SPECIFICATION

COATING, SILVER CONDUCTIVE, ELECTRICAL AND THERMAL

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of silver-filled, electrically conductive coating.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

TT-P-143	Paint, Varnish, Lacquer, and Related Materials, Packaging, Packing, and Marking of
PPP-C-96	Cans, Metal, 28 Gage or Lighter

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

FSC 8010

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MIL-STD-129 Marking for Shipment and Storage

MIL-STD-130 Identification Marking of U.S. Military Property

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Uniform Classification Committee

Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60607.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified, a preproduction sample shall comply with all the requirements specified herein.

3.2 Materials. The coating material shall consist of a silver-filled organic resin and a volatile solvent.

3.3 Curing characteristics. The coating shall cure by solvent evaporation under the following temperature conditions (see 4.9.2.2):

a. Plus 23 degrees plus or minus 5 degrees Celsius (C) (plus 73 degrees plus or minus 9 degrees Fahrenheit (F)) for 24 hours.

b. Plus 90 degrees plus or minus 10 degrees (plus 194 degrees plus or minus 18 degrees F) for 1 hour).

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3.4 Resistance. The resistance of the coating shall be not more than 1.0 ohm per square per mil when measured after curing and conditioning coated test panels at the temperatures specified in 4.9.2.2.

3.5 Workmanship. The components shall be mixed and processed in such a way as to ensure compliance with the requirements of this specification. Freshly-opened containers of the coating material shall be free from grit, sediment and suspended matter. The coating material shall be smooth and homogeneous.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspections of the coating are classified as follows:

a. Preproduction sample inspection, see 4.4.

b. Quality conformance inspection, see 4.5.

c. Preparation for delivery inspection, see 4.6.

4.3 Inspection lot. For purposes of sampling, an inspection lot shall consist of a single batch of coating material offered for inspection at one time. Each batch shall have been produced in a single vat in no more than a 24-hour continuous production period with no change in process or raw materials.

4.4 Preproduction sample inspection. Unless otherwise specified, a sample of the coating material from the first lot manufactured under each contract or order shall be subjected to the preproduction inspection. The preproduction sample shall be taken from a lot produced by the same processes as are proposed for all the coating material furnished under the same contract or order.

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4.4.1 Preproduction sampling and testing. One container of coating material shall be selected at random from the preproduction lot, and samples therefrom shall be subjected to tests specified in Table I in the order listed.

Table I

Preproduction Inspection		
Examination or test	Requirement paragraph	Inspection method paragraph
Visual examination	3.5	4.9.1
Resistance	3.4	4.9.2

4.4.2 Action in case of failure. If a failure occurs during preproduction inspection, the lot shall be rejected. The supplier shall take corrective action as necessary to preclude recurrence of the failure in subsequently-manufactured batches. Acceptance of subsequently-manufactured batches may be withheld until the corrective action has been taken and a batch which incorporates the corrective action has been subjected to and has passed preproduction inspection.

4.5 Quality conformance inspection. All production lots of the coating material offered for acceptance hereunder shall be subjected to the quality conformance inspection. One container of coating material shall be selected at random from the inspection lot and shall be subjected to the tests in Table I in the order listed. If a failure occurs, the lot shall be rejected.

4.6 Preparation for delivery inspection. Each lot offered for acceptance shall be subjected to the preparation for delivery inspection.

4.6.1 Sampling and testing. Samples shall be taken in accordance with the applicable specifications and standards referenced in 5.1, 5.2, and 5.3. When no sampling plan is specified by those specifications and standards, samples shall be taken in accordance with MIL-STD-105, acceptable quality level (AQL) 2.5. The samples should be inspected as specified in 4.9.3.

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4.6.2 Action in case of failures. Unless other criteria are defined in the applicable documents referenced in 5.1 through 5.3 inclusive, failure of one specimen shall be charged as a single defective. If the quality of defectives exceeds that permitted, the lot represented by the sample shall be rejected. Defective specimens shall not be delivered on contract or purchase order until all deficiencies have been corrected.

4.7 Inspection equipment and facilities. Inspection equipment shall be of adequate quality and quantity to permit performance of the required examinations and tests, and shall comply with the equipment described in the applicable inspection methods of this specification and the documents referred to thereby. Inspection facilities shall be adequate to permit performance of the required inspection.

4.8 Standard inspection conditions. Unless otherwise specified, all inspections shall be performed under conditions of room ambient pressure, temperature, and humidity.

4.9 Inspection methods.

4.9.1 Visual examination. The coating material in a freshly opened container shall be visually examined for compliance with the requirements specified in 3.5. If necessary, small quantities of the coating material shall be poured onto clean glass surfaces for examination.

4.9.2 Resistance test.

4.9.2.1 Preparation of test panels. Six test panels shall be prepared by coating glass slides with a strip of the coating material 0.0005 to 0.0010 by 1/16 by 6 inches.

4.9.2.2 Cure and conditioning. Three of the test panels shall be cured at plus 23 degrees plus or minus 5 degrees C (plus 73 degrees plus or minus 9 degrees F) for 24 hours; these shall be referred to as Type A panels. The other three panels shall be cured at plus 90 degrees plus or minus 10 degrees C (plus 194 degrees plus or minus 18 degrees F) for 1 hour; these shall be referred to as Type B panels. Immediately after curing, the panels shall be conditioned in accordance with Table II.

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Table II

Conditioning for resistance test		
Panel type and quantity	Temperature	Duration
1A 7 1B	+74° + 6°C (+165° ± 10°F)	1 hour, -0, +6 minutes
1A & 1B	-54° + 3°C (-65° ± 5°F)	1 hour, -0, +6 minutes
1A & 1B	-54° ± 3°C	72 hours, minimum*

* After exposure to -54°C for 72 hours, these two panels shall be stabilized at room temperature prior to measuring the resistance.

4.9.2.3 Procedure. With the exception noted in Table II, 3 volts direct current (dc) shall be applied across the strip of coating on each test panel immediately after conditioning. The resistance shall be read with an ohmmeter, and the value recorded in ohms per square per mil, using the following conversion:

Resistance (ohms) x Thickness (mils)/96 squares = Resistance (ohms/square/mil)

4.9.3 Preparation for delivery. Inspection shall be performed to determine whether the material is prepared for delivery in accordance with the applicable requirements of 5.1, 5.2, and 5.3.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level A. The coating shall be furnished in metal containers conforming to PPP-C-96. The size of the containers shall be as specified in the contract or purchase order. Containers shall be provided with wire handles which have been galvanized or protectively coated to resist corrosion.

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5.1.2 Level C. Packaging shall be done in accordance with the supplier's commercial practice, and shall provide adequate protection against deterioration during shipment from the supply source to the first receiving activity.

5.2 Packing.

5.2.1 Level A. The containers shall be packed for overseas shipment in compliance with TT-P-143, Level A.

5.2.2 Level B. The containers shall be packed in compliance with the domestic shipping requirements of TT-P-143, Level B.

5.2.3 Level C. The containers shall be packed for shipment in compliance with Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Identification. Containers of the coating shall be marked in compliance with MIL-STD-130. The following information shall appear on each container:

a. Title, number, and date of this specification.

b. Manufacturer's name or trademark.

c. Manufacturer's formula number.

d. Batch number.

e. Date of manufacture.

f. Quantity.

g. Recommended thinner and instructions for use.

5.3.2 Marking for shipment. Unit containers and exterior shipping containers shall be marked in compliance with MIL-STD-129.

6. NOTES

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6.1 Intended use. The coating material covered by this specification is intended for use in making electrical connections in guided missile circuits.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether preproduction inspection is required.

c. Size of container.

d. Required warranty of shelf life, if any. Terms such as the following are suggested:

"The supplier shall warrant that the coating material will meet all requirements hereof for one year of storage in the unopened container under ambient storage conditions, and that he will replace any container which fails to meet all requirements hereof during such term of storage."

e. Selection of applicable levels of packaging and packing.

6.3 Thinner. The supplier will recommend an appropriate thinner which will restore the coating material to spraying consistency in the event of solvent evaporation. (See 5.3.1)

6.4 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-13098, dated 13 April 1965.

Custodian:
Army-MI

Preparing Activity:
Army-MI

Project No. 8010-A089

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

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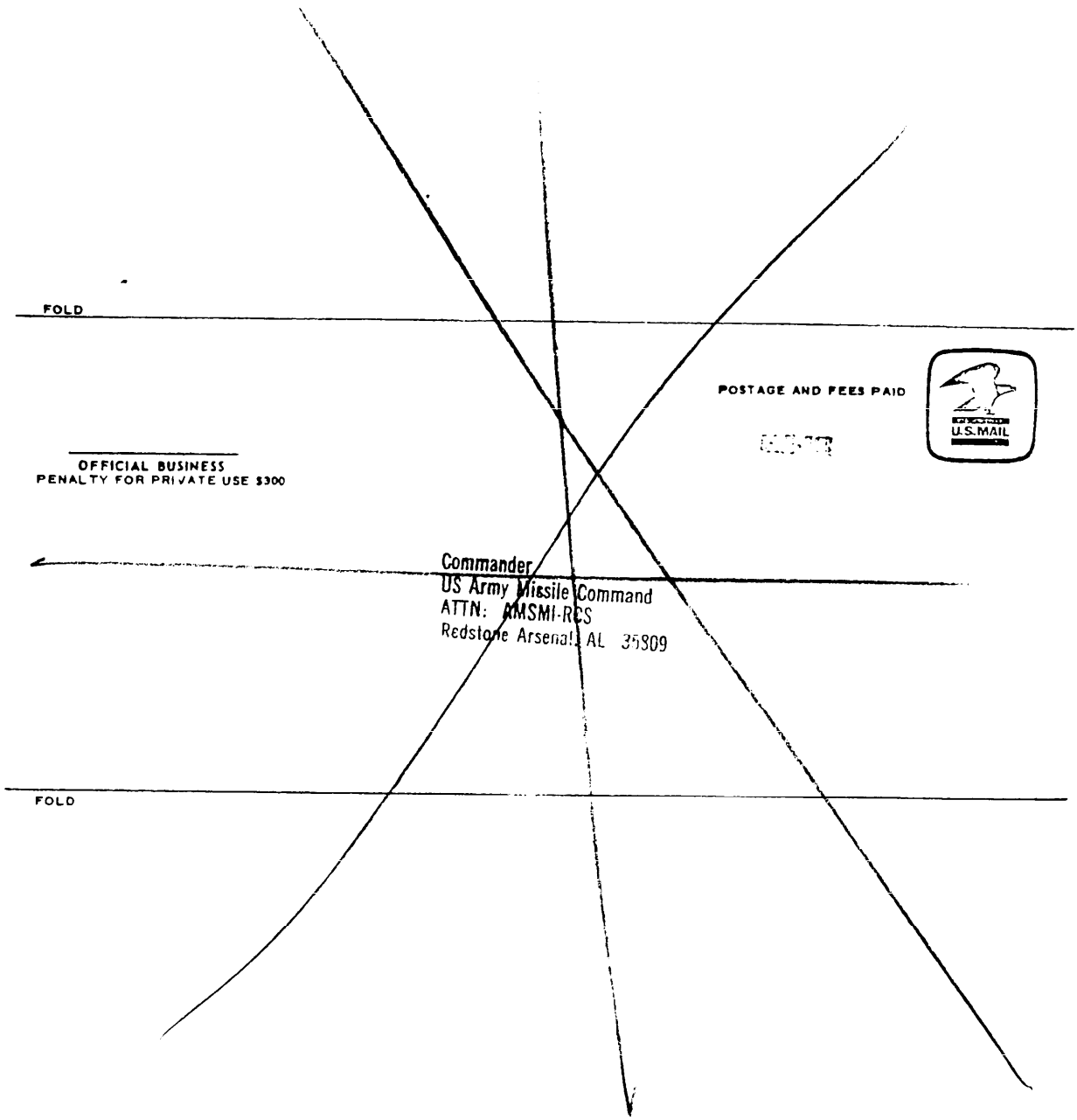
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